Hall Ticket Number:

MR-21

Code: B314CS5S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July/August-2024

Discrete Mathematics

(Common to CSE, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

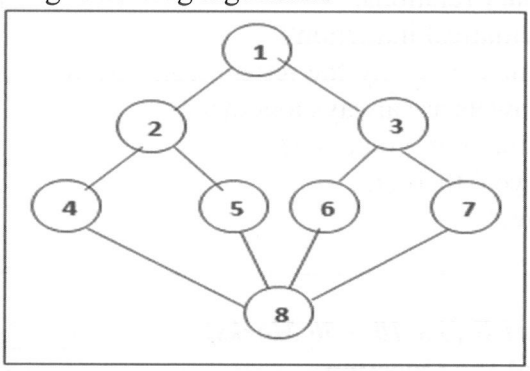
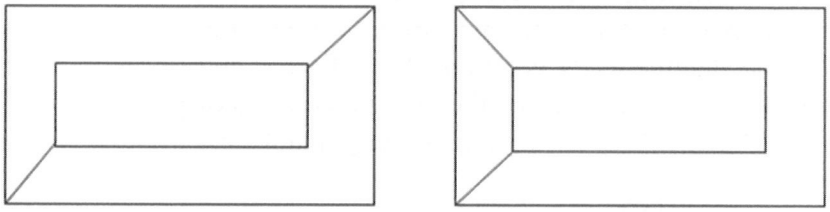
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Construct the truth table of the compound proposition $(p \vee q) \rightarrow (p \wedge q)$.	2	3	1	1
1. b)	What do you mean by well-formed formula, give example?	2	1	1	1
1. c)	Define the term POSET, give example.	2	3	2	1
1. d)	Elaborate the properties of binary relations.	2	1	2	1
1. e)	What is the principle of Mathematical induction?	2	1	3	1
1. f)	How many words can be formed by using the letters from the word "DRIVER" such that all the vowels are always together?	2	3	3	1,2
1. g)	What is the probability of getting 3 or 5 on a die?	2	3	4	1
1. h)	List out the applications of Bayes' theorem.	2	2	4	1
1. i)	What are the applications of tree?	2	2	5	1
1. j)	Define eulers formula.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.	5	1	1	1,3
2. b)	Consider these statements "All lions are fierce", "Some lions do not drink coffee", "Some fierce creatures do not drink coffee" Let P(x),Q(x), and R(x) be the statements "x is a lion", "x is fierce" and "x drinks coffee" respectively. Assuming that the domain consists of all creatures express the statement in the argument using quantifiers and P(x),Q(x) and R(x).	5	3	1	1,2
	OR				
2. c)	Show that $r \wedge (p \vee q)$ is a valid conclusion from the premises $(p \vee q)$, $(q \rightarrow r)$,$(p \rightarrow m)$ and $(\sim m)$.	5	1	1	1,2
2. d)	"If there was a ball game, then traveling was difficult. If they arrived on time, then traveling was not difficult. They arrived on time. Therefore, there was no ball game." Show that these statements constitute a valid argument.	5	4	1	1,3
3. a)	Consider the following relation on {1,2,3,4,5,6}, $R = \{(i,j): \lvert i - j \rvert = 2\}$ Is R transitive? Is R reflexive?	5	2	2	1
3. b)	Prove that a relation R on A is symmetric if and only if $R = R^{-1}$	5	2	2	1
	OR				
3. c)	Draw the Hasse diagram representing the partial ordering. {(a,b) / a divides b} on {1,2,3,4,6,8,12} .	5	1	2	1,2
3. d)	Prove that $A-(B \cap C) = (A-B) \cup (A-C)$.	5	3	2	1
4. a)	Use mathematical induction to prove that $1 + 2 + 3 + \ldots + n = n(n+1)/2$ for all positive integers n.	5	3	3	1,2
4. b)	State Pigeon hole principle. Make use of it, find how many people were born on the same month among 200 people.	5	3	3	1,2
	OR				

P.T.O.

4. c)	Use mathematical induction to show that $1 + 2 + 2^2 + 3^3 + \ldots\ldots + 2^n = 2^{n-1}$	5	3	3	1,2
4. d)	Compute the number of rows of 6 Americans, 7 Mexicans and 10 Canadians in which an American invariably stands between a Mexican and a Canadian and in which a Mexican and a Canadian never stand side by side.	5	3	3	1,3
5. a)	Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions	5	2	4	1,2
5. b)	Let $X \sim Uniform(a,b)$. Find $E(X)$.	5	2	4	1
	OR				
5. c)	A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn and are found to be both hearts. Find the probability of the lost card being a heart.	5	3	4	1,2
5. d)	Solve the recurrence relation of the Fibonacci series of numbers.	5	2	4	1
6. a)	Show that K_5 is an Euler's circuit and also Hamiltonian cycle.	5	2	5	1
6. b)	Explain BFS using following algorithm. 	5	2	5	1,3
	OR				
6. c)	Verify whether two graphs G1 and G2 are isomorphic or not. 	5	2	5	1,2,3
6. d)	State and prove fundamental theorem of graph theory.	5	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome